B2.2 Organelles and compartmentalization

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b2-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 248–255 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

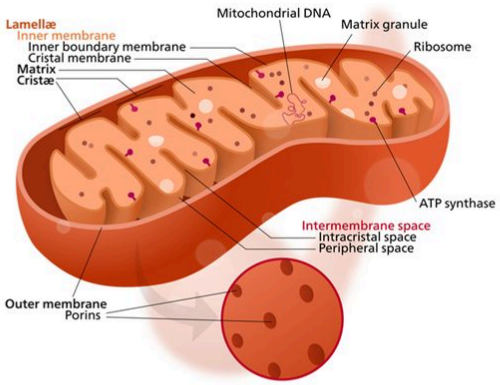
Organelles that make ATP

B2.2.4 The mitochondrion

pp. 249–250

The mitochondrion: built to make ATP

B2.2.4



Mitochondrion. Kelvinsong, modified by Sowlos, CC BY-SA 3.0.

Structure	Adaptation
Double membrane	Separates respiration from cytoplasm
Small intermembrane space	Steep proton gradient fast
Cristae	Large area for ETC and ATP synthase
Matrix	Krebs cycle enzymes concentrated

Key idea
Compartments let different conditions sit side by side.

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A **double membrane** separates aerobic respiration from the cytoplasm. The inner membrane is folded into **cristae**, giving a large surface area for electron transport chains and ATP synthase. Protons are pumped into the **small intermembrane space**, so a steep gradient builds quickly. The **matrix** holds the Krebs cycle enzymes and substrates together at a suitable pH.

B2.2.5 The chloroplast

pp. 250–252

Stacked **thylakoid membranes** (grana) give a large surface area for photosystems, so much light is absorbed. Protons pumped into the **small volume** inside the thylakoids quickly make a steep gradient for ATP synthase. The **stroma** holds the Calvin cycle enzymes, such as rubisco, and their substrates, separate from the light-dependent reactions.

B2.2.6 The double nuclear membrane

pp. 252–253

The two membranes of the nuclear envelope keep DNA apart from the cytoplasm. They join at the **nuclear pores**, which control traffic: mRNA leaves; nucleotides, proteins and ATP enter. The outer membrane is continuous with the rough ER. At the start of mitosis and meiosis the envelope breaks into **vesicles**, so spindle microtubules can reach the chromosomes; afterwards the vesicles re-form a nucleus around each set.

Making and moving proteins

B2.2.7 Free ribosomes and rough ER

pp. 253–254

Both are 80S ribosomes. **Free ribosomes** make proteins that stay in the cytoplasm, or are imported into the nucleus, mitochondria or chloroplasts. Ribosomes bound to the **rough ER** make proteins for secretion, for lysosomes or for the plasma membrane; these enter the ER lumen and travel on in vesicles. Several ribosomes on one mRNA (a polysome) make many copies at once.

B2.2.8 The Golgi apparatus

pp. 254–255

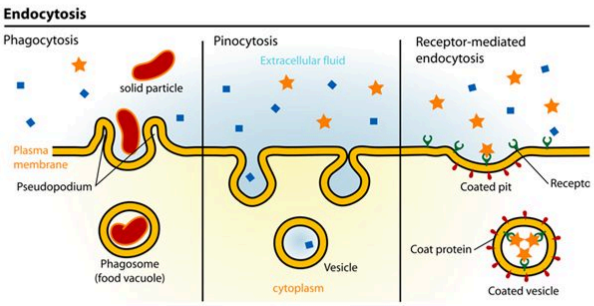
A stack of flattened sacs (**cisternae**). Vesicles from the rough ER fuse with the side facing the ER. Proteins pass from cisterna to cisterna, where enzymes modify them, for example by adding sugars (glycosylation) or phosphate. These changes label proteins for their destination. Vesicles bud from the far side and carry the proteins to the plasma membrane for secretion, or to lysosomes.

B2.2.9 Vesicles and clathrin

Vesicles and clathrin

B2.2.9

Substance binds receptors → **Clathrin cage forms inside** → Membrane pulled into a pit → Pit pinches off → Coat removed; vesicle moves



Phagocytosis, pinocytosis, receptor-mediated endocytosis. M. Ruiz Villarreal, public domain.

Uses

Cells take in LDL cholesterol, iron and hormones this way; viruses exploit it too.

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Vesicles are small sacs of membrane that carry materials between compartments and across the plasma membrane. In **receptor-mediated endocytosis**, a substance binds receptors and **clathrin** proteins link into a cage on the inner face of the membrane, pulling it into a coated pit that pinches off as a vesicle. Cells take in LDL cholesterol this way, and many viruses use it to enter.

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